

Generation Interconnection System Impact Study Report

For Queue Project Q20-1418

Carmi 345 kV
March 06, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1418. The Interconnection Customer proposes a 379.8 MW solar photovoltaic facility interconnecting at the Carmi 345 kV substation in the Maryland area. The request was received on March 04, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 535 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,286,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MILL CTEEK - COLORA 138 KV	CARMi - BIXBY HILL 345 KV L	125.0	229	5.2
LAMBIE - KIMBERLY CARK 138 KV	BIXBY HILL - CARMi SUBSTATION 345 KV	116.6	272	5.2
ROCK SPRINGS SUBSTATION - CARMi 138 KV line	BIXBY HILL - CARMi 345 KV line	109.3	326	5.2
COLWOOD TAP - BIG CRK 138 KV	BIXBY HILL - CARMi 345 KV Line	109.3	313	5.2
ROCK SPRINGS SUBSTATION - BIG CRK 138 KV line	BIXBY HILL - CARMi 345 KV L	108.8	320	5.2
COLWOOD TAP - LAMBIE 138 KV	BIXBY HILL - CARMi 345 KV	106.8	311	5.2
COLORA - KIMBERLY CLARK 138 KV Line	BIXBY HILL - CARMi 345 KV Line	105.3	284	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MILL CTEEK - COLORA 138 KV	Rebuild line with higher-capacity conductor	\$6,809,000	62% (\$4,227,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
KIMBERLY CLARK - LAMBIE 138 KV	Rebuild line with higher-capacity conductor	\$3,371,000	62% (\$2,093,000)
ROCK SPRINGS - CARMi 138 KV	Reconductor line	\$673,000	62% (\$418,000)
COLWOOD TAP - BIG CRK 138 KV L	Reconductor line	\$640,000	62% (\$397,000)
ROCK SPRINGS - BIG CRK 138 KV	Reconductor line	\$2,806,000	62% (\$1,742,000)
COLWOOD TAP - LAMBIE SUBSTATION 138 KV	Reconductor line	\$1,351,000	62% (\$839,000)
COLORA - KIMBERLY CLARK 138 KV Line	Reconductor line	\$919,000	62% (\$570,000)
	Total allocated to project		\$10,286,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.